



COMPUTER SECURITY

2025

FINAL 2020 IS
(Solved)



SENIOR
25

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Choose the correct answer:

1.	Type of firewall can decide what services to perform on the user's behalf in accordance with its available knowledge			
	a) Screening routers	b) Stateful inspection	c) Guards	d) Application proxies
2.	It perform a simple pattern-matching to report situations that corresponding to a known attack type			
	a) Normal based	b) Signature-based IDS	c) Heuristic IDS	d) Honey pot
3.	It implies overhearing <u>without expending any extra effort</u>			
	a) Eavesdrop	b) Wiretap	c) Spoofing	d) Session hijacking
4.	Interception communications <u>through some effort</u> is known as _____ attack			
	a) Session hijacking	b) Spoofing	c) Wiretap	d) Eavesdrop
5.	Consider case1, as shown in table one, the suitable firewall type is _____			
	a) Application proxies	b) Stateful inspection	c) Screening routers	d) Guards
6.	Triple-DES with two keys id defined by $C = E_{k3}(D_{k2}(E_{k1}(P)))$, if $K1$ equal $K3$, the triple-DES can be considered			
	a) Double DES	b) Triple DES	c) Single DES	d) Useless encryption
7.	Which of the following isn't a design goal of firewalls?			
	a) Always invoked	b) Tamper-proof	c) Complex code	d) Simple functionality
8.	It tracks the sequence of packets and conditions from one packet to another to thwart an attack			
	a) Screening routers	b) Stateful inspection	c) Application proxies	d) Guards
9.	Encrypting $P = \text{"stay home to be safe"}$ by block transposition algorithm $k = 32514$, then C is Stayh ometo besaf exxxx لو $k = 32514$ يبقى نستنتج ان: اول حرف هيبقى ثالث واحد في التشفير تاني حرف هيبقى ثاني حرف ثالث حرف هيبقى خامس حرف رابع حرف هيبقى اول حرف خامس حرف هيبقى رابع حرف Ciphertext: athsy emoot sefba xxxex			
	a) SOBETMEWAESXYT AJHOFZ	b) AESXTMEWHOFZSO BEYTAJ	c) UOFZYTAJAESXTME WSOBD	d) None
10.	A sender <u>should not be able falsely to deny later that he/she sent a message</u>			
	a) Integrity	b) Confidentially	c) Privacy	d) Non-repudiation
11.	Consider case 2, as shown in table 1 , the suitable firewall type is _____			
	a) Screening routers	b) Stateful inspection	c) Application proxies	d) Guards
12.	Consider case 3 , as shown in table 1, the suitable firewall type is _____			
	a) Personal firewall	b) Stateful inspection	c) Application proxies	d) Guards



13.	Using the play fair matrix shown in fig 1 , the cipher of “FACULTY” will be _____						M	F	H	I/J	K
	FA CU LT YI HL DQ AS XK						U	N	Q	P	Q
							Z	V	W	X	Y
							E	L	A	R	G
							D	S	T	B	C
							a)	HLDQSAKX	b)	HDQASLXK	c)
14.	Physical OS separation approach _____ the resource sharing and _____ the security level										
	a)	Minimizes , Maximizes		b)	Maximizes , Maximizes		c)	Maximizes , Minimizes		d)	Minimizes , Minimizes
15.	Kerberos system is considered _____ party _____way authentication technique										
	a)	Two , One		b)	Two , Two		c)	Three , Two		d)	Three , One
16.	Digital signature provides the following characteristics except _____										
	a)	Message authentication		b)	Receiver authentication		c)	Sender authentication		d)	Message encryption
17.	_____ Is used for managing audit trails and highlighting user violation of policy or normal activity										
	a)	Firewall		b)	IDS		c)	Antivirus		d)	Virus scanner
18.	Correction system configuration error is a function of _____										
	a)	Heuristic-based IDS		b)	Personal firewall		c)	Proxy		d)	Signature-based IDS
19.	Denning’s IDS model builds heuristic decision model based on tracing _____										
	a)	Actions and it’s initiator		b)	Timestamp of each action		c)	Exception condition of each action		d)	All
20.	The result of encoding a plaintext “COMPUTER” based on play fire cipher with key “PARTY” , consider(J,Z) are less used characters CO MP UT ER DN UB						P	A	R	T	Y
							B	C	D	E	F
							G	H	I/J	K	L
							M	N	O	Q	S
							U	V	W	X	Z
	a)	DNGTNTAT		b)	DNTBWATD		c)	NDGTAWTA		d)	DGWNTADT
21.	Consider the cipher text obtained from Q20, encrypting it with row transposition algorithm, K=2, results _____										
	a)	DGWDNTAT		b)	DGWDTATN		c)	NGATDTWA		d)	DWTGDNAT
22.	Based on the play fair matrix of Q20, the plaintext of the cipher text “NXOMTW” Is _____										
	a)	Hellox		b)	Sunset		c)	Party		d)	None
23.	Cyber security rules for daily practicing are classified as:										
	a)	Hardware controls		b)	Software controls		c)	Administration controls		d)	Physical controls
24.	_____ means that some unauthorized party (person, program) has gained access to data										
	a)	Interception		b)	Modification		c)	Interruption		d)	Fabrication
25.	Legal and ethical controls are classified as:										
	a)	Hardware controls		b)	Software controls		c)	Policy controls		d)	Physical controls



26.	A virus is designed to hide itself from detection by antivirus software							
	a)	Brain virus	b)	Boot sector virus	c)	Polymorphic virus	d)	Trojan horse
27.	_____ record an intruder's actions and even attempt to trace who the attacker is from actions, packet data, or connection							
	a)	ISS scanner	b)	Nessus	c)	Honey pot	d)	tripwire
28.	Which of the following is/are considered a candidate IDS profile?							
	a)	Logon and session activity	b)	Command or program usage	c)	File access activity	d)	All
29.	_____ used to protect the networks hosts from an outside address							
	a)	Firewall	b)	IDS	c)	Antivirus	d)	Virus scanner
30.	Denning's IDS model is example of _____							
	a)	Host based IDS	b)	Signature based IDS	c)	Anomaly based IDS	d)	Hybrid IDS
31.	In which different processes use different physical objects, such as separate printers for output requiring different levels of security							
	a)	Temporal separation	b)	Logical separation	c)	Physical separation	d)	Cryptographic separation
32.	Cryptography is used as a defense technique against the following attacks except _____							
	a)	Modification	b)	Interception	c)	Interruption	d)	Fabrication
33.	Flooding is an attack of _____							
	a)	Integrity	b)	Availability	c)	Confidentiality	d)	All are true
34.	Virus is active as long as the computer running							
	a)	Memory resident	b)	Brain virus	c)	Polymorphic virus	d)	Trojan horse
35.	Program controls include:							
	a)	Peer review	b)	Proof of program correctness	c)	Independent testing	d)	All

Case 1	A library wants to make a available certain documents but, to support fair use of copyrighted matter, it will allow a user to retrieve only the first so many characters of a document, after that amount, the library will require the user to pay a fee that will be forwarded to the author
Case 2	A university wants to allow its students to use e-mail up to a limit of so many messages or so many characters of e-mail in the last so many days, although this result could be achieved by modifying e-mail handlers, it is more easily done by monitoring the common point through which all e-mail flows, the e-mail transfer protocol
Case 3	A company wants to allow dial- in access by its employees, without exposing its company resources to login attacks from remote non-employees

